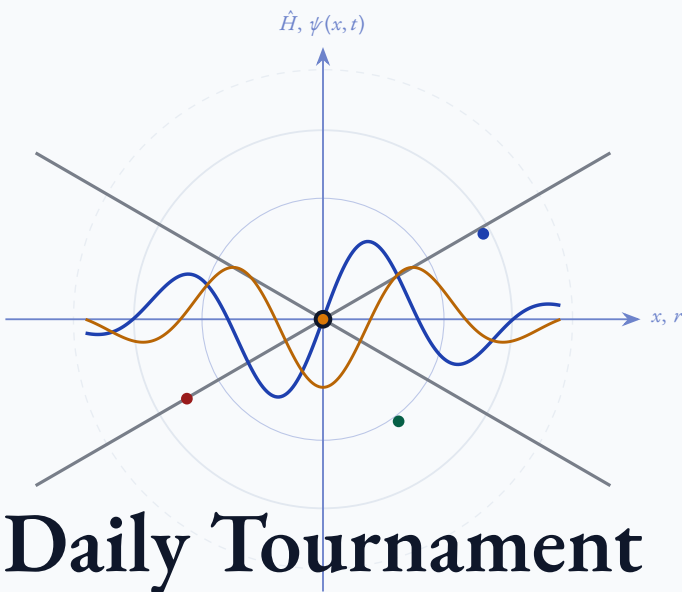




Daily Tournament

GRAND ARENA OF THEORETICAL PHYSICS · v6.0

“Today’s contestants: [604 Core Focus] × [811 Core Focus]”
Closed-book · Pen-and-paper derivation · Muscle memory reconstruction

Date	[YYYY-MM-DD] Week [X] (Day [N]) Level [L]: [Rank Title] ([XP] XP)
Modules	604: [Math Topics] 811: [QM Topics]
Architecture	v6.0 Deluxe: Muscle Matrix (Blue) → Concept Blitz (Amber) → Core Model (Green) → Abyss (Wine)
Target	Full 6-Page Deluxe Combat Edition Standard 80 min Complete Run

Theoretical Physics Training • Daily Tournament

[YYYY-MM-DD] | Week [X] (Day [N]) | Phase [P]: [Phase Name]

§1 Tactical Briefing & Warm-Up Retrieval

- 604 今日靶向:[阐述今日 604 高数的核心计算攻坚目标、重点公式与避坑原则]。
- 811 今日靶向:[阐述今日 811 量子力学的物理模型、算符代数或概念直觉目标]。
- Spaced Retrieval 间隔提取 (Day +3 滚动回溯): 604 (Day 3 回溯) [3 天前易错数学公式]; 811 (Day 3 回溯) [3 天前易错物理公式]。
- 考试规则: 闭卷、独立手推、严禁查阅笔记; 先在下方草稿区完成 5-8 分钟白纸公式破冰与间隔提取。

Warm-Up Retrieval Checklist (5-8 min): without notes, write on scratch paper:

☐ [604 Focus 1] [核心公式 1] ☐ [604 Focus 2] [核心公式 2] ☐ [811 Focus 1] [核心公式 3] ☐ [811 Focus 2] [核心公式 4]

Warm-Up Retrieval Workspace | 白纸破冰与间隔提取草稿区

5-8 分钟闭卷盲写: 默写今日核心公式与边界条件, 快速越过大脑初始静摩擦力。

§2 Foundation Matrix & Concept Blitz

Complete independently under timed conditions

§A 基础肌肉记忆区 (Muscle Memory Matrix · 30 min)

Problem A.1 | 604 Calculus: [Problem Type]

Suggested time: 10 min

[此处填入 604 微积分基础计算题:如泰勒极限展开、定积分计算、反常积分审敛等。必须写清详细步骤要求。]

Problem A.2 | 604 Linear Algebra: [Problem Type]

Suggested time: 10 min

[此处填入 604 线性代数基础计算题:如行列式按行展开、矩阵初等变换、特征值与特征向量求解等。]

Problem A.3 | 604 ODE / Advanced Calculus: [Problem Type]

Suggested time: 10 min

[此处填入 604 常微分方程或基础多元微积分题目:如一阶线性方程、齐次/非齐次常系数方程等。]

§B 量子概念闪击战 (Quantum Concept Blitz · 10 min)

Problem B.1 | 811 Concept: [Topic Name] (图像辨析)

Suggested time: 5 min

[此处填入判断题或快问快答:考察波函数意义、测量公理、不同图像(Schrödinger vs Heisenberg)等物理概念。]

Problem B.2 | 811 Concept: [Topic Name] (守恒律与量纲)

Suggested time: 5 min

[此处填入量纲分析、守恒量与对称性分析、或物理量概率流解释题目。]

§3 Core Model Mastery & The Abyss

The Backbone of UCAS Exam

§C 核心母题攻坚 (Core Model Mastery • 25 min)

■ Problem C.1 | 811 Model: [Core Physics Model]

Suggested time: 25 min

[此处填入 811 量子力学核心标准母题。必须设置引导性小问 (1), (2), (3), 按梯度推进: 哈密顿量 → 本征态/对易方程 → 物理讨论。]

或矩阵元构建 ;

求解或微扰展开 ;

趋势或极限对应 。

§D 极限挑战区 (The Abyss • 选做 • 15 min)

■ Problem D.1 | Advanced Exam Trap: [Boss Model]

Optional • Suggested time: 15 min

[此处填入高危超纲真题变式或压轴题: 如二重积分极坐标高斯型积分、Virial 定理、算符指数 BCH 展开等。标注为选做。]

§4 Model Solutions & Commentary

Complete your independent attempt on scratch paper before consulting this page.

Section A · 604 Foundation Solutions

A.1: [写出严密推导与最终化简结果。]

[Common Error] [指出该题常见易错点、未定式陷阱或符号错误。]

A.2: [写出特征值计算与自检验证(迹与行列式)。]

A.3: [写出一阶线性常微分方程通解推导。]

Section B & Section C · 811 QM Solutions

B.1: [概念正误判断与一句话物理原理解释。]

B.2: [概率流密度或守恒律的物理图像白话阐述。]

C.1: (1) [小问 1 推导与结果];

(2) [小问 2 推导与结果];

(3) [小问 3 推导与物理意义]。

[Physical Insight] [总结该题体现的核心量子原理或 Ehrenfest 对应定理。]

Section D · Abyss Problem Solution

[写出选做题的完整推导过程,指出巧妙的换元或几何转化思想。]

§5 Daily Debrief & Reflection (XP Checkout)

Post-Match Review (fill in before sleep, 5 min)

Section A (肌肉记忆) Time: _____ min | A.1 ☐ | A.2 ☐ | A.3 ☐ | 全对 (+30 XP) ☐

错因剖析: ☐ 泰勒漏阶 ☐ 特征多项式计算失误 ☐ 积分因子公式遗忘

Section B (概念闪击) Time: _____ min | B.1 ☐ | B.2 ☐ | 概念直觉清晰 (+10 XP) ☐

错因剖析: ☐ 图像定义颠倒 ☐ 连续性方程物理意义不清晰

Section C (核心母题) Time: _____ min | 独立推导出核心方程 (+50 XP) ☐ | 看了解析 ☐

错因剖析: ☐ 对易子计算符号错误 ☐ 运动方程定义模糊

Section D (深渊挑战) Time: _____ min | 成功攻破压轴大题 (+20 XP 选做成就) ☐ | 未作答 ☐

Total XP Check 今日结算经验值: _____ XP | 当前累计: [Base XP] + _____ XP

State Assessment ☐ Peak performance (冲级状态) ☐ Steady progress (平稳推进) ☐ Minimum defence mode (保底防御)

Key Derivation Insights & Traps | 突破口与避坑沉淀

Core Breakthrough (今日核心突破口与关键一步):

Fatal Trap & Common Error (致命陷阱与避坑警示):

Day +3 Action Item (3 天后间隔重做提取的具体题号与断点):